

Cavity-induced persistent current in a π loop

Model. We consider a superconducting (SC) loop interrupted by a single Josephson element and threaded by the magnetic flux of one lumped LC mode [Fig. ??(a)],

$$H = \hbar\omega a^\dagger a + H_{\text{ring}}[\varphi_{\text{cl}} + g(a + a^\dagger)]. \quad (1)$$

$H_{\text{ring}}[\varphi]$ is a microscopic lattice Hamiltonian of the loop: an s -wave BCS chain $-t \sum_{r\sigma} (c_{r+1\sigma}^\dagger c_{r\sigma} + \text{h.c.}) + \Delta \sum_r (c_{r\uparrow}^\dagger c_{r\downarrow}^\dagger + \text{h.c.})$ closed by a junction, which is either a weak bond t_J or an Anderson impurity (level ϵ_d , Hubbard repulsion U) tunnel-coupled to both banks. The total Peierls phase on the loop is placed on one junction bond, $-t_J \sum_\sigma e^{i\varphi} c_{1\sigma}^\dagger c_{0\sigma} + \text{h.c.}$. An electron phase $\varphi = \pi$ corresponds to one superconducting flux quantum $\Phi_0 = h/2e$. The vector potential of the mode enters through the gauge-invariant displacement operator $D = e^{ig(a+a^\dagger)}$, so diamagnetic (A^2) terms are included to all orders. $g = 2\pi\Phi_{\text{zpf}}/\Phi_0^{(e)}$ is the electron phase generated by the zero-point flux of the mode, and φ_{cl} is a static external flux. For a lumped resonator of impedance Z , galvanically sharing its inductance with the loop, $(2g)^2 = \pi Z/R_Q$ with $R_Q = h/4e^2$.

The only property of H_{ring} that enters below is its ground-state energy at fixed phase,

$$E(\varphi) = \sum_{m \geq 0} E_m \cos m\varphi. \quad (2)$$

Even harmonics ($m = 2, 4, \dots$) describe Cooper-pair tunnelling ($h/2e$ periodicity). The $m = 1$ term is the single-electron (h/e) persistent current of a mesoscopic ring [1] and is suppressed as $e^{-L/\xi}$. A 0-junction has $E_2 < 0$ and a π -junction has $E_2 > 0$. When a 0-junction is biased by $\Phi_0/2$, $\varphi_{\text{cl}} = \pi/2$ maps $E_2 \cos 2\varphi \rightarrow -E_2 \cos 2\varphi$. The two routes to a π loop are therefore equivalent up to the odd harmonics, which turn into an explicit symmetry-breaking term $-E_1 \sin \varphi$. In contrast, an intrinsic π -junction keeps $E(\varphi) = E(-\varphi)$ exactly, since $H_{\text{ring}}[0]$ is real (time-reversal symmetry).

Adiabatic reduction and Landau theory. For $\hbar\omega$ well below the Andreev gap E_A , the electrons follow the flux adiabatically. To leading order in $\hbar\omega/E_A$, Eq. (1) reduces to a single-particle problem for the dimensionless flux $X = a + a^\dagger$,

$$H_{\text{BO}} = \frac{\hbar\omega}{4} (P^2 + X^2) + E(\varphi_{\text{cl}} + gX), \quad [X, P] = 2i. \quad (3)$$

Equation (3) is the Hamiltonian of an rf-SQUID [2] whose inductive energy is supplied by the mode. Writing it in the Cooper-pair phase $\theta = 2gX$ gives $4E_C n^2 + \frac{1}{2}E_L \theta^2 + E(\theta/2)$ with $E_L = \hbar\omega/8g^2$ and $E_C = \hbar\omega g^2$. A static field ($\langle a \rangle \neq 0$), i.e. a photon condensate carrying a persistent current $I = -(2e/\hbar)\partial_\varphi E$, appears classically when the

orbital response of the ring at the symmetric point is paramagnetic and exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2 |E''(\varphi_0)|}{\hbar\omega} > 1, \quad E''(\varphi_0) < 0. \quad (4)$$

For a pure π -junction, $\beta = 8g^2 E_J/\hbar\omega = 2\pi L I_c/\Phi_0$, the familiar screening parameter of the loop [3, 4]. Equation (4) is the single-loop form of the photon-condensation criterion of Andolina *et al.* [5]: the no-go theorem for a uniform cavity field [6, 7] does not apply, because the mode couples to the flux (curl of $\mathbf{A}$), and the instability requires a *paramagnetic* orbital response, which here comes from the π shift. Above threshold, the Landau free energy $\mathcal{F}(X) = \hbar\omega X^2/4 + E(gX)$ has two degenerate minima $\pm X^*$ carrying opposite currents $\pm I^*$.

Two remarks fix the physical content of Eq. (4). First, for a loop with its own geometric inductance L_r and mutual inductance $M \leq \sqrt{L_r L_c}$ to the resonator, minimizing the static energy over the resonator flux gives $\Phi_r^2/2L_r$: the static threshold is set by the total loop inductance, and Eq. (1) is the $k = M/\sqrt{L_r L_c} \rightarrow 1$ (galvanic) limit. Second, zero-point fluctuations of the mode do not drive the transition; they oppose it. At finite $E_J/\hbar\omega$ the ground state of Eq. (3) is the symmetric superposition of the two current-carrying states, split from its partner by a tunnelling gap $\delta \sim \hbar\omega e^{-S}$ with an instanton action $S \propto (E_J/\hbar\omega) s(\beta)$. A sharp quantum phase transition with mean-field exponents is recovered in the scaling limit $E_J/\hbar\omega \rightarrow \infty$ at fixed β , in which $E_C/E_J = \beta(\hbar\omega)^2/(8E_J^2) \rightarrow 0$. This is the analogue of the classical-oscillator limit of the quantum Rabi model [8].

Methods. We solve Eq. (1) without approximation by DMRG (TeNPy [9]) on the full electron-photon Hilbert space. The ring is stored in folded order next to a bosonic site (Fock cutoff N_{ph}). Fermion parity and S^z are conserved, and the electron-photon vertex is a three-site operator $D c_{1\sigma}^\dagger c_{0\sigma}$. Benchmarks against exact BdG energies of the non-interacting ring at $g = 0$ agree to 10^{-14} at full bond dimension ($\chi = 1024$). In production ($\chi = 256-512$) the energy error is below 3×10^{-6} and the error in the current $\langle I \rangle$ below 10^{-7} [check]. Excited states are obtained by DMRG orthogonal to the ground state. We compare with (i) the self-consistent electron $\times$ photon product state (mean field, MF), and (ii) the adiabatic Hamiltonian (3), with $E(\varphi)$ from BdG or from DMRG at fixed φ , diagonalized exactly.

Breakdown of the product-state description. We first take the parameters of a Zeeman-polarized dot embedded in a short BCS ring ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$). The self-consistent product state predicts a current-carrying phase for $g \gtrsim 0.75$, with $\langle a^\dagger a \rangle \simeq 1-1.4$ and $|\langle I \rangle| \approx 0.02$

[Fig. 2]. DMRG shows that this phase is an artefact. The exact ground-state energy decreases monotonically with g and lies below the MF energy by up to $1.9\hbar\omega$ [update with DMRG $g=1.2$]. The gap to the first excited state stays a sizeable fraction of $\hbar\omega$ ($E_1 - E_0 = 0.34\hbar\omega$ at $g = 1.2$). The linear response to the seed flux $\varphi_{cl} = 10^{-3}$ remains $\langle I \rangle \lesssim 4 \times 10^{-4}$, i.e. two orders of magnitude below the MF value. DMRG and the adiabatic theory (3) agree to within $\lesssim 2\%$ of $\hbar\omega$ in energy and to $< 1\%$ in $E_1 - E_0$. The residual difference is the non-adiabatic (diagonal Born–Oppenheimer) correction $\sim g^2\hbar\omega\hbar\omega/E_A$.

Two effects make the product state fail. (i) It is not gauge invariant [10, 11]: with the full Peierls phase on one bond, it renormalizes that bond by $|\langle D \rangle| < 1$ and so loses phase-independent kinetic energy, which the exact state recovers by following the flux adiabatically. (ii) Here $E_J \simeq \hbar\omega$. The Landau free energy does develop a double well for $g > 0.58$, but its depth is below the zero-point energy, and the flux distribution $P(X)$ stays essentially unimodal. A further subtlety concerns ring size: since $L \lesssim \xi$, $E(\varphi)$ of this ring is dominated by the single-electron harmonic, $E_1 = 2.1 \times 10^{-2} \gg E_2 = 2.0 \times 10^{-3}$ [Fig. 1(a)]. The instability would then be one of the normal h/e persistent current rather than of the Josephson junction. To isolate Josephson physics we use rings with $L \gg \xi$ below ($t = 1$, $\Delta = 1$, $L = 14$, giving $|E_1/E_2| \lesssim 10^{-2}$).

Half-flux-biased junction. For an ordinary weak link ($t_J = 0.5$, $E_2 = -0.056t$, Andreev gap $0.77t$) biased by $\Phi_0/2$, Fig. 3 shows the full DMRG solution at $E_J/\hbar\omega = 10$ as a function of β . [fill: DMRG current jumps to $|\langle I \rangle| \simeq I^*$ at $\beta \simeq 1$, follows BO within ..., MF ...] The flux distribution extracted from the photon reduced density matrix evolves from a Gaussian centred at zero to a single displaced peak at $\pm\Phi^*$. The sign is fixed by the residual odd harmonic $E_1 = -5.4 \times 10^{-4}$, which tilts the double well by $2|E_1| \sin \varphi^* \approx 10^{-3}t \approx 0.2\hbar\omega$. This tilt sets the splitting $E_1 - E_0$ for $\beta > 1.2$ in DMRG and BO alike. Without it, $\delta/\hbar\omega$ would drop below 10^{-6} at $\beta = 2$. The external-bias route thus provides a current-carrying state whose direction is *selected* rather than spontaneously chosen. Its residual symmetry breaking is $\propto e^{-L/\xi}$ for a homogeneous ring, but in practice it is limited by the flux offset $\delta\Phi$.

Interaction-driven π junction without external flux. We now replace the weak link by an Anderson impurity at particle-hole symmetry ($\epsilon_d = -U/2$, $t_J = 0.5$) and set $\varphi_{cl} = 0$. DMRG in the even and odd fermion-parity sectors [Fig. 1(b,c)] gives the $0-\pi$ transition known for magnetic quantum dots [12–18]. For $U < U_c \simeq 0.9t$ the singlet (even) state is the ground state at $\varphi = 0$ and the junction is a 0 -junction. For $U > U_c$ the Kramers doublet (odd sector) is the ground state at all phases, and $E(\varphi)$ has its *minimum* at the Cooper-pair phase π . At $U = 2t$ we find $E_2 = +7.39 \times 10^{-3}t$, with $|E_1|, |E_4| < 10^{-4}t$: a pure π junction, indistinguishable after normalization

from the half-flux-biased weak link [Fig. 1(a)]. The π coupling decreases monotonically with U ($E_2 \propto U^{-1.3}$ for $4t \leq U \leq 10t$), as expected for cotunnelling through the singly occupied level.

Coupling this junction to the mode [Fig. 4] gives the transition without any static flux. Here H_{ring} is real and $\langle I \rangle = 0$ in every eigenstate, so the order is diagnosed by the two lowest states. The splitting $\delta = E_1 - E_0$ collapses and the flux operator projected onto the quasi-degenerate doublet acquires the eigenvalues $\pm\Phi^*$. [fill DMRG numbers: at $E_J/\hbar\omega = 10$, $\delta/\hbar\omega = \dots$ at $\beta = 2$ and ... at $\beta = 3$; Φ^* ...]. In the photon-number basis the DMRG ground state is a cat state: $P(\Phi)$ is bimodal with peaks at $\pm\Phi^*$, whose separation tracks the classical minimum. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [Fig. 4(b)]. The residual δ is the macroscopic tunnelling of a persistent current. Because it is protected by time-reversal symmetry, it is the only energy scale separating the two current directions. Each current state is additionally Kramers degenerate in the spin of the dot. The low-energy manifold is therefore a fourfold multiplet $\{\sigma = \uparrow, \downarrow\} \otimes \{\odot, \ominus\}$, split only by δ .

Experimental scales. The two control parameters are $\beta = 2\pi LI_c/\Phi_0$ and $E_J/\hbar\omega$, related by $\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$, with $E_J/h \simeq 0.5 \text{ GHz} \times I_c[\text{nA}]$. Three classes of π elements span the phase diagram [Fig. ??]: (i) SFS junctions ($I_c \sim 1\text{--}100 \mu\text{A}$) reach $\beta > 1$ with geometric inductances of a few to a few hundred pH and $E_J/\hbar\omega \sim 10^2\text{--}10^4$. This is the classical limit, where the half-flux state is fully polarized, consistent with the spontaneous currents observed in SFS and d -wave π loops [4, 19, 20]. (ii) Quantum-dot π junctions in the doublet regime ($I_c^\pi \sim 0.1\text{--}10 \text{ nA}$) require $L \sim 0.03\text{--}3 \mu\text{H}$, i.e. superinductances (Josephson-junction arrays, granular Al) as in fluxonium circuits [21, 22]; this is the π -qubit regime of Refs. [23–25]. These give $E_J/\hbar\omega \sim 1$: the ground state is the cat state $(|\odot\rangle + |\ominus\rangle)/\sqrt{2}$ with $\delta \sim 0.1\hbar\omega$, and the transition appears as a dip in the $|0\rangle \rightarrow |1\rangle$ frequency and a jump in the transition matrix element $\langle 0|\Phi|1\rangle$, accessible in two-tone spectroscopy. (iii) Multichannel or gate-tunable π junctions with $I_c \sim 50\text{--}200 \text{ nA}$ in a high-kinetic-inductance loop ($L \sim 2\text{--}10 \text{ nH}$, $\omega/2\pi \sim 2\text{--}5 \text{ GHz}$) give $\beta \sim 1\text{--}3$ and $E_J/\hbar\omega \sim 10\text{--}30$. This is the regime of Figs. 3 and 4: a well-defined current-carrying state whose residual tunnelling $\delta/\hbar\omega \sim 10^{-4}\text{--}10^{-8}$ remains a measurable macroscopic-quantum-tunnelling signature. In the externally biased realization, the half-flux bias must be stable to $|\delta\Phi| \ll \delta/2I^*$, i.e. $\lesssim 10^{-6}\Phi_0$ for the parameters above. The interaction-driven realization has no such requirement.

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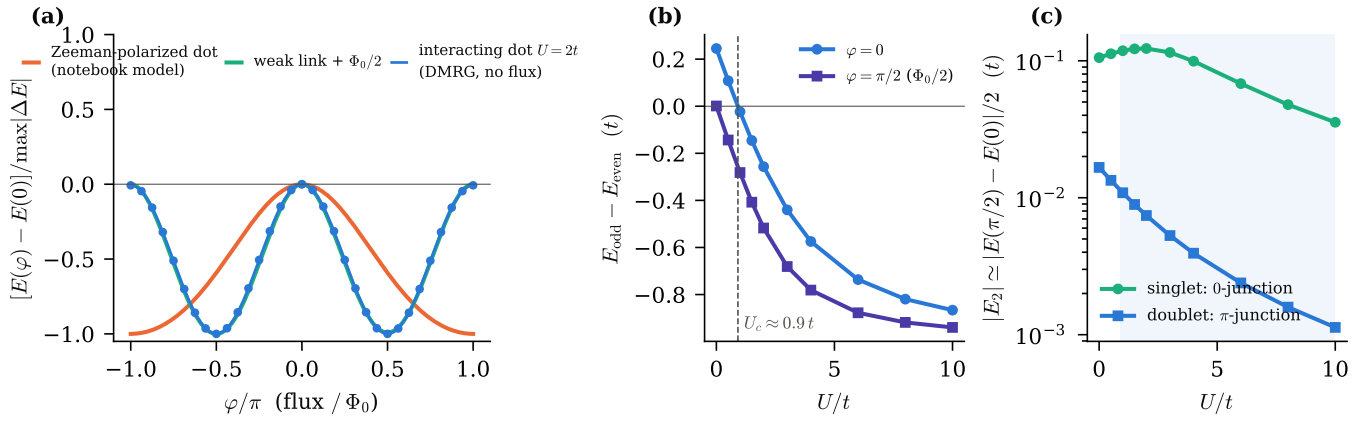


FIG. 1. (a) Normalized ground-state energy of the loop versus the electron phase φ (flux in units of $h/2e$) for three junctions: Zeeman-polarized dot in a short ring (dominated by the single-electron harmonic $\cos \varphi$), an ordinary weak link biased by $\Phi_0/2$, and an interacting-dot ring at $U = 2t$ without any bias (DMRG, symbols; line: cosine fit). (b) Parity competition $E_{\text{odd}} - E_{\text{even}}$ of the interacting-dot ring at $\varphi = 0$ and $\varphi = \pi/2$ (DMRG, $L = 14$, $\Delta = t$, $t_J = 0.5t$). (c) Josephson energy $|E_2|$ in the singlet (0) and doublet (π) sectors. Shading: doublet ground state.

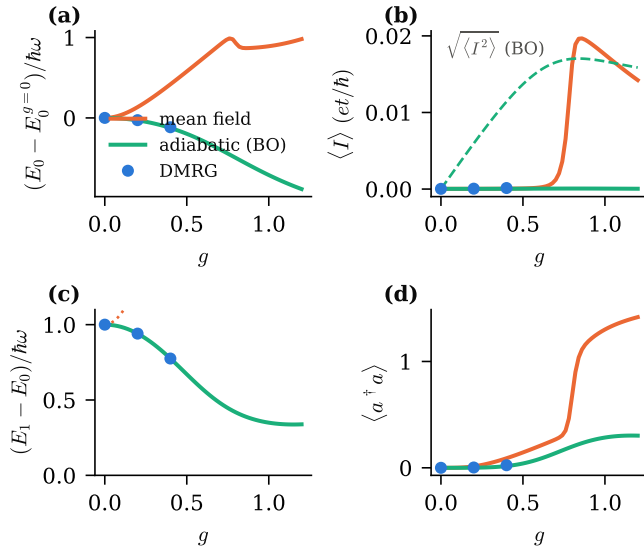


FIG. 2. Short ring with a Zeeman-polarized dot ($\hbar\omega = 0.02$, $\varphi_{\text{cl}} = 10^{-3}$): mean field (MF, orange), adiabatic theory (3) (green), and DMRG (symbols). (a) Ground-state energy, (b) current (dashed: rms current of the BO ground state), (c) gap to the first excited state, (d) photon number.

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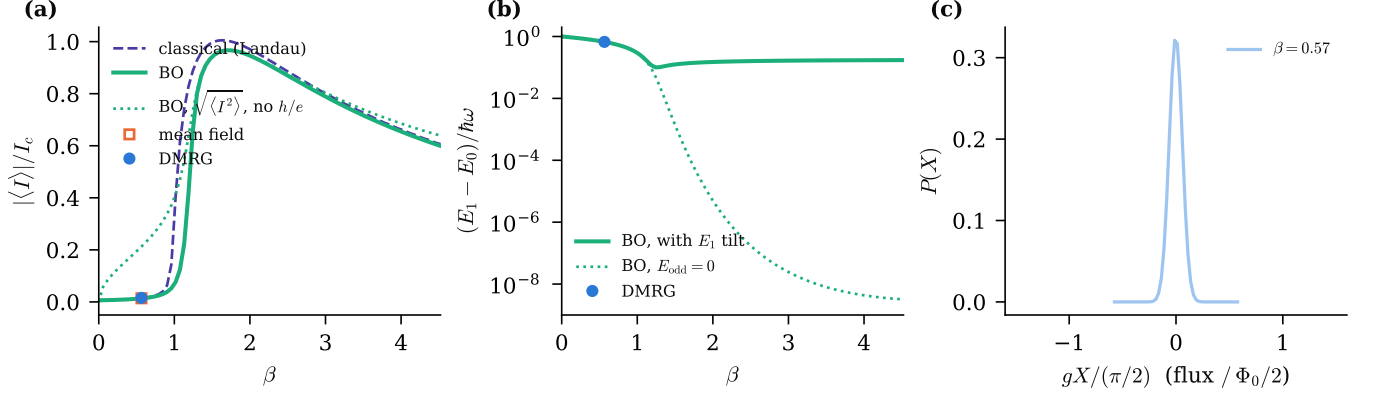


FIG. 3. Weak-link loop biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current versus β : DMRG (circles), MF (squares), adiabatic theory with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state, obtained from the photon reduced density matrix.

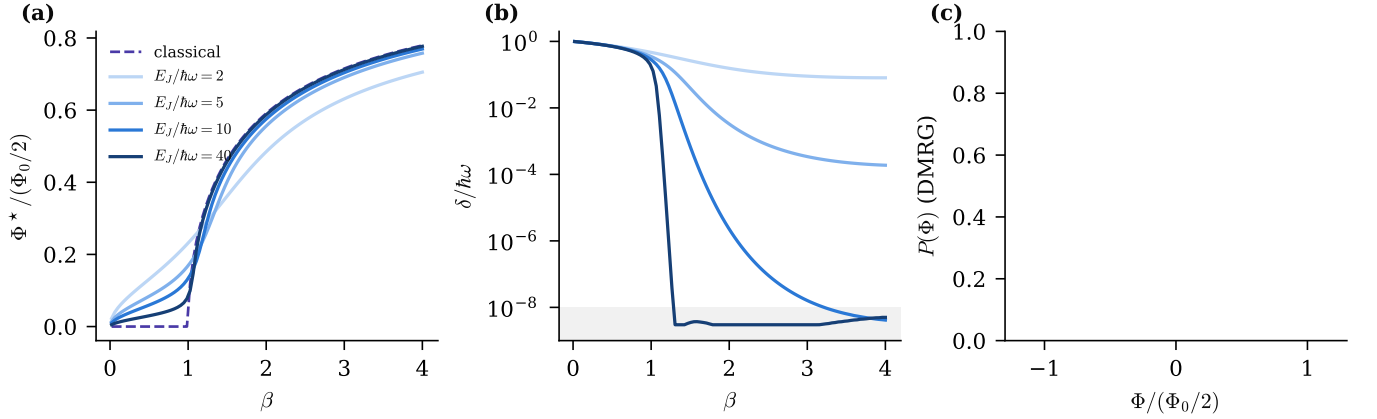


FIG. 4. Interaction-driven π loop ($U = 2t$, no external flux). (a) Flux of the condensate Φ^* (eigenvalue of the flux projected on the two lowest states) versus β for increasing $E_J/\hbar\omega$ (lines: adiabatic theory with the DMRG $E(\varphi)$; circles: full DMRG at $E_J/\hbar\omega = 10$; dashed: classical). (b) Tunnel splitting δ between the two current-carrying states. (c) Flux distribution of the DMRG ground state (a cat state for $\beta > 1$).